

RAG Indexing Overview

Retrieval-Augmented Generation (RAG) grounds an LLM in an external document collection. The pipeline has three stages: chunking, embedding and indexing.

Chunking splits documents into smaller passages. Fixed-size chunking uses uniform windows with overlap; it is simple and predictable but can cut sentences and sections in the middle. Structural chunking splits by document structure - markdown headings, code functions, paragraphs - producing semantically coherent chunks with meaningful section labels.

Embeddings map each chunk to a dense vector so that semantically similar passages are close in vector space. Cosine similarity ranks chunks by relevance to a query embedding.

Metadata attached to each chunk (source, file, section, chunk_id) lets the system cite the origin of retrieved passages and filter by document.

The index can be stored in FAISS, SQLite or JSON. For small corpora a brute-force cosine search over a SQLite-backed vector store is enough.